MARITIME RISK INTELLIGENCE

Wake
AI

Making dark shipping visible.

From signal gaps
to actionable cases.

PENN BUILT

SEPT 2026

01 / THE PROBLEM

When a vessel goes dark, the trail goes cold.

Vessels can disable AIS—the system that broadcasts their location, speed, and heading.

That blackout can hide sanctioned oil movements and ship-to-ship cargo transfers.

THE INVESTIGATOR SEES A GAP—NOT THE ACTIVITY INSIDE IT.

Not a fringe problem.

558	**~25%**	**7.8M**
average suspected dark tankers	of the global crude tanker fleet	metric tons of crude moved per month

43% of officially recorded global seaborne crude exports in the study comparison

One signal is never enough.

AIS gaps

When & where a vessel disappears

Geography

Risky routes & sanctioned-port proximity

Behavior

Speed, heading & voyage anomalies

Proximity

Potential ship-to-ship transfers

KEY PRINCIPLE An AIS gap is not proof. Context turns a gap into a lead.

04 / THE PRODUCT

Wake AI finds the signals worth waking up to.

× DETECT × CONNECT × EXPLAIN × PRIORITIZE

05 / LIVE DEMO

Watch a blackout become a case.

01 Normal voyage begins

02 AIS signal disappears

03 Wake AI finds correlated risk signals

04 Analyst receives the evidence-backed case

06 / ARCHITECTURE

How Wake AI works.

01	02	03	04
Data ingestion	**Event engine**	**Risk model**	**Analyst workflow**
AIS + vessel + geospatial data	Gaps, routes, and proximity	Signals become explainable scores	Alerts, cases, and evidence

DETAIL TO BE ADDED: data sources, model logic, and analyst interface.

WAKE AI

When ships go dark, risk should not.

Make maritime activity visible, explainable, and actionable.

BUILT BY A PENN TEAM · INSPIRED BY DARK-SHIPPING RESEARCH COAUTHORED BY PENN PROFESSOR JESÚS FERNÁNDEZ-VILLAVERDE